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BLUETOOTH CONTROL Through Mobile

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Creating a Bluetooth-controlled device with an Arduino Uno and an HC-05 Bluetooth module enables control of lights or even relays to manage AC appliances wirelessly using an Android smartphone.

Communication between the smartphone and the Arduino Uno is facilitated by the HC-05 Bluetooth module, which provides an easy and cost-effective method to integrate remote control functionality into various devices. By utilising MIT App Inventor, it is possible to develop a custom app that sends commands to the Arduino, allowing complete control over connected devices, such as toggling lights on or off.

Whether controlling an LED or an AC device, this setup demonstrates a simple yet powerful application of Arduino and Bluetooth technology. The author's prototype is shown in Fig. 1. The list of components required for the device

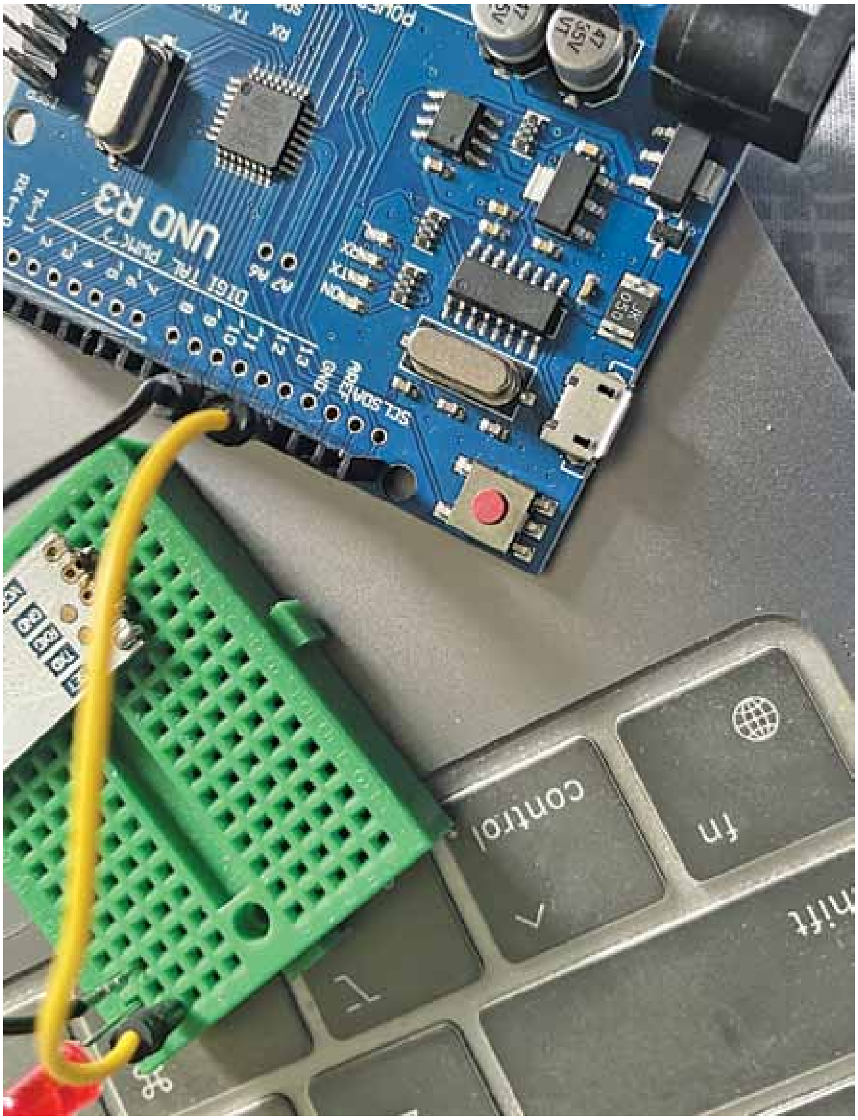

Fig. 1: Author's prototype

BILL OF MATERIALS

Item Name	Quantity
Arduino Uno with USB cable	1
HC-05 Bluetooth module	1
Breadboard	1
Resistor 1-kilo-ohm (R3)	1
Resistor 2-kilo-ohm (R2)	1
Resistor 330-ohm (R1)	
5mm LED	1
Connecting wires	As required

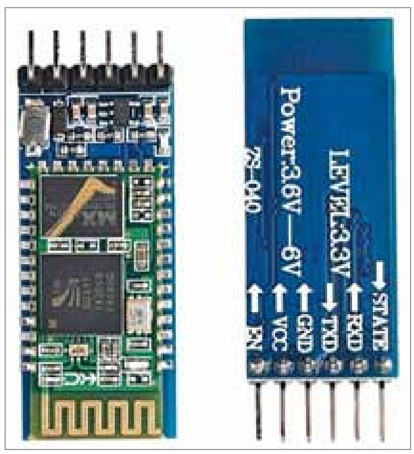

Fig. 2: HC-05 Bluetooth module